Database	
- Stores user - Stores campsites - Stores notification log - Stores waitlist - Stores reservation list - Interacts with the Firebase backend by querying and storing information - Stores messages - Stores lotteries	- Firebase -

Device Model	
- Stores user geolocation - Store user credentials - Identifies user by deviceID	- UserModel - Map

Abstract Class: User Model	
- Special roles (entrants, owners, admins) - Verify if user exists already - Make a new user, which it adds to the database - Check if user has saved credential bypasses login screen - Remove user / delete profile	- Database - DeviceModel

Current user information (login, pfp)

AccountController	
- Reads input given by LoginView or SignupView - Ensure input is query-ready by applying regex to clean it (trimming whitespace/rejecting empty searches) - Gives query-ready input to **User** to match results	- UserModel - DeviceModel

LoginView	
- Text boxes: username, password -	- UserModel - DeviceModel

SignupView	
- Text boxes: Create password, confirm password, username -	- UserModel - DeviceModel

Class: CampsiteOwner Extends User Model	
- Create and manage campsite(s) - Upload (add/replace) promo images - Attach geolocation of campsite - Monitor the progress of the waitlist - Access to final list of reservees	- CampsiteModel - LotteryController - CalendarModel - WaitlistModel - CampsiteView - ReservationModel

Class: Entrant Extends User Model	
- store profile picture - Remove profile picture - Tracks if user wants to have notification opt-out	- Database - DeviceModel

Class: Administrator Extends User Model	
- Create admin accounts - Send fire forest warnings - be able to be a campsite owner and/or entrant with my admin profile.	- NotificationModel - AdministratorView - UserModel

CampsiteModel	
- Update campsite information - Store campsite ratings - Remove campsite - Store and Remove promo image - Store and create waitlist - Store real-time status updates - Generate QR code - Store campsite rules and policies - Two modes: continuous booking vs fixed set of dates	- GeolocationModel - CampsiteOwnerModel - EntrantModel - CampsiteView - Map - LotteryController - ReservationModel - QRcode

- Rating -

SearchBarController	
- Reads input given by SearchBarView - Ensure input is query-ready by applying regex to clean it (trimming whitespace/rejecting empty searches) - Gives query-ready input to FilterBy to match results	- SearchBarView - FilterBy

SearchBarView	
- Search button & text input boxes for location and dates - Display results list - Send whatever the user inputs in the text box to SearchBarController to handle - Update view when notified by FilterBy	- SearchBarController - FilterBy

FilterBy Model	
- Get filtered results by querying the database with input given by SearchBarController - Notifies the view when results are available by calling update()	- SearchBarController - SearchBarView

- Location
- Date
- Number of guest

CalendarModel	
- Track availability per campsite -	- CampsiteModel - ReservationModel

EntrantsView	
- Display filtered campsites based on location, date, and number of guests -	- SearchBarView - SearchBarController - FilterBy

CampsiteSummaryView	
	- CampsiteModel

- Name - Location - Maximum occupancy - rating

CampsiteInfoView	
- Display button to join active lottery - Display details of campsite - Display button to join lottery	- WaitlistModel - CampsiteModel

WaitlistModel	
- Add and remove entrants to list - Stops adding entrants when full	- CampsiteModel - EntrantModel - Database

WaitlistView	
- Button to leave waitlist	- WaitlistModel

NotificationView	
- Displays list of recent notification -	- LotteryLoseView - LotteryWinView - ConfirmationView

LotteryLoseView	
- Display lose message with campsite name	- CampsiteModel

LotteryWinView	
- Display win message with campsite name - Display available dates of the campsite that can be reserved for - Button to accept reservation - Button to decline reservation - Checkbox for entrants to confirm that they have read the campsite rules and policies before confirming their reservation	- ReservationModel - CampsiteModel - LotteryModel

- available_dates

ReservationModel	
- Track which days are booked by which entrants - Stores reservation status (pending, accepted/declined, canceled/confirmed) - Tracks if the user has paid/unpaid, or been refunded - Update final reserve list for campsiteowner - Allow campsite owner to reject reservation	- EntrantModel - CampsiteModel - Database - CampsiteOwnerModel

- List<Date>
- reservation_list

LotteryController	
- draw/chooses people from waitlist after registration period ends - Creates reservation for winners - Sends win and lose notification - Manages when winning entrant accepts or decline	- WaitlistModel - ReservationModel - NotificationView - LotteryWinView - LotteryLoseView - EntrantModel

ConfirmationView	
- Displays confirmation message to entrant of campsite reservation	- campsiteModel - EntrantModel

ActiveReservationView	
- Display entrants active reservations - Button to cancel reservation	- ReservationModel - EntrantModel - ReservationController

PastReservationView	
- Displays past reservations	- ReservationModel

ProfileInfoView	
- Button to Upload photo - Display delete profile button - Toggle to opt out of notifications	- EntrantModel -

- Name
- Email
- Phone number
- profile

PolicyView	
- Display app policy and guidelines	

QRScan	

RegisterCampsiteView	
- campsite details including name, location, description (amenities), maximum occupancy, map location, optional image, and tags. - Text boxes to input those buttons	- CampsiteOwner - CampsiteModel

CampsiteOwnerView	
- Displays lists of campsite owners campsites - Open to show details of campsites - Display the currently booked dates for my campsite.	- CampsiteOwner - CampsiteModel - CampsiteInfoView

CreateLotteryView	
- list multiple campsites in a lottery for a certain period. - Set registration period - Set booking period - Set number of lottery winners - optionally set a capacity for my lottery waitlist. - button to post lottery	- LotteryModel - CampsiteModel

EditLotteryView	
- Display current lottery information that can be editable - Save button	- LotteryModel

LotteryModel	
- store registration period - store booking period - store number of lottery winners - Store optionally set a capacity for my lottery waitlist. - Update lotterylist - automatically draw winners after the registration period ends.	- CampsiteModel - Waitlist - LotteryController - Database

CampsiteOwnerReservationView	
- Display a a list of active reservations. - Display a list of past reservations - Display a list of canceled reservations - Button to reject a reservation	- ReservationModel

CampsiteOwnerWaitlistView	
- Display entrants on the waitlist for a specific campsite - Button to notify entrants who have joined the waitlist for my campsite lottery.	- WaitlistModel - NotificationModel

NotificationModel	
- Stores notifications (specific to user) - Stores long of sent notifications for campsiteowner	- Database - EntrantModel - CampsiteOwnerModel - Waitlist

MessageModel	
- Stores messages between campsiteowner and entrant - Allow campsite owner to send message to entrant	- Database - ReservationModel

MessageView	
- Displays messages between campsite owner and entrant - Textbox for messages - Back button - Button to send message - Display profile picture and name of recipient	- messagemodel

AdministratorView	
- Browse and remove registered campsites - Browse and Remove profile - Browse and Remove images - remove campsite owners that violate app policy. - review logs of all notifications. - review logs of all messages from campsite owners to entrants who reserved their campsite.	- UserModel - NotificationModel - MessageModel - CampsiteModel
